

# Krishna P Tambatkar

+91-9972696546 | [ktams2530@gmail.com](mailto:ktams2530@gmail.com) | [krishnapt.is-a.dev](https://krishnapt.is-a.dev) | [github.com/krishh2530](https://github.com/krishh2530) | [linkedin.com/in/krishnatambatkar](https://linkedin.com/in/krishnatambatkar)

## PROFILE

B.Tech graduate in Electronics and Computer Engineering. Hands-on experience building embedded IoT systems – secure cellular connectivity, RTOS-based firmware, and cloud integration. Aspiring software engineer skilled in Python, C, and machine learning, with a track record of measurable, high-impact results.

## SKILLS

**Technical Skills:** C, Python, Embedded Systems/RTOS, MQTT, TLS, AWS (IoT Core, Lambda), TensorFlow, PyTorch, Data Structures & Algorithms, Machine Learning  
**Soft Skills:** Leadership, Team Player, Communication, Versatility

## EXPERIENCE

### Wipro

*IoT Software Engineering Intern*

*Apr 2026 – Present*

- Designed and implemented solutions on Quectel EC800G cellular module (Embedded C, RTOS) using dedicated RX/TX threads to establish secure 4G LTE TCP/IP sockets, routing bi-directional telemetry to AWS IoT Core.
- Configured TLS 1.2 encryption and SNI routing; provisioned and authenticated AWS X.509 certificates directly on device flash memory.
- Engineered a full-duplex MQTT client with QoS 1 message delivery and C-level memory buffering to parse cloud downlinks and prevent UART overflow.
- Diagnosed and resolved hardware-level control-flow and network packet failures between the edge device and AWS servers.

### Samsung R&D Institute

*Software Engineering Intern*

*Nov 2025 – Apr 2026*

- Designed a hybrid 5G receiver combining a pretrained neural network with a classical detection algorithm, achieving a **3.41 dB performance gain** over the existing baseline, without retraining either model.
- Derived the optimal weight for combining both methods through statistical analysis, validated with strong statistical confidence across 1,000+ test cases.
- Outperformed the previous best neural network approach using **45% fewer parameters** and zero additional training time, while achieving real-time inference speed (**1.8 ms**).
- Built the complete simulation and benchmarking pipeline using NVIDIA Sionna and TensorFlow.

## PROJECTS

### Blockchain-based Safety Alert System for Internet of Vehicles (IoV)

Decentralized System

- Built a decentralized real-time risk management system to broadcast critical events (drowsy, drunk, rash driving, accidents) using Ethereum smart contracts and a React Native (Expo) infotainment app.
- Integrated a blockchain relay server with AWS Lambda for scalable cloud-based event pipelines, ensuring secure, low-latency, and reliable communication.

### Menstrual Phase-Specific Dietary Guidance

SPIN 2025

- Developed a system providing customized dietary suggestions by predicting menstrual phases with **99.39% precision** using XGBoost and SMOTE.
- Published and presented research in the 12th International Conference SPIN 2025.

## EDUCATION

### Amrita Vishwa Vidyapeetham

Bengaluru, India

*B.Tech in Electronics and Computer Engineering (CGPA: 7.23/10.0)*

*2022 – 2026*

## LEADERSHIP & ACTIVITIES

### Team Lead, AMAL Club (Amrita Management and Leadership)

2024 – 2025

- Led an **8+ member team** to organize a Mock Parliament with **40+ participants**, ensuring 100% on-time delivery of logistics and task execution.

### Sponsorship Team, College Fest

2023

- Secured sponsorship deals from **15+ local businesses** through outreach, brochure presentations, and stakeholder engagement.